Hints: Linear algebra – 3. Eigenvalues, diagonalisability and the Jordan form

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. A real matrix with only real eigenvalues can be triangularised by an orthogonal matrix (real Schur form): $A = QTQ^T$ with T upper triangular. Rewrite the equation in terms of T.

Hint 2. $T + T^k = T^T$: the left side is upper triangular and the right side is lower triangular, so T is diagonal. Then $T^k = O$ for a diagonal T. Answer: only $A = O$.

Problem 2

Hint 1. $A^m = 2021B - B^2$ is symmetric, and its eigenvalues are $2021\beta - \beta^2 \leqslant 2021^2/4$. Use this for even exponents.

Hint 2. $A^{2m} = (A^m)^2$ is the square of a symmetric matrix, so its eigenvalues lie in $[0, 2021^2/4]$. Then $|\det A|^{2m} = \det A^{2m} \leqslant (2021^2/4)^n$ for every m; let $m \to \infty$.

Problem 3

Hint 1. Compute A^4 using $A^2 = A^T$ twice.

Hint 2. $A^4 = (A^T)^2 = (A^2)^T = A$, so every eigenvalue satisfies $\lambda^4 = \lambda$: $\lambda = 0$ or $\lambda^3 = 1$. All four values occur: O, E, and the 2×2 rotation by $2\pi/3$ (for which $R^T = R^{-1} = R^2$). Answer: $\lambda \in \{0, 1, e^{\pm 2\pi i/3}\}$.

Problem 4

Hint 1. Write the rotation as a matrix product: with J the anti-diagonal permutation matrix (ones on the anti-diagonal), $A^R = JA^T$. So the condition is $A = JA^T$, i.e. $A^T = JA$, and transposing, $A^T = AJ$.

Hint 2. Hence $JA = AJ$, and $A^TA = JA^2 = AA^T$: A is normal. For a normal real matrix, $Av = \lambda v$ implies $A^Tv = \bar{\lambda}v$.

Hint 3. Then $\bar{\lambda}v = A^Tv = JAv = \lambda Jv$. If $\lambda \neq 0$, v is an eigenvector of J with eigenvalue $\bar{\lambda}/\lambda$, and $J^2 = E$ forces $(\bar{\lambda}/\lambda)^2 = 1$, i.e. $\bar{\lambda} = \pm\lambda$.

Problem 5

Hint 1. Factor $3x^3 - x^2 - x - 1 = (x - 1)(3x^2 + 2x + 1)$.

Hint 2. The roots are 1 and two complex roots of modulus $1/\sqrt{3}$, all simple. So A is diagonalisable over $\mathbb{C}$ with eigenvalues among these. Hence A^k converges to the projection onto the 1-eigenspace along the others, which is idempotent (and real, as a limit of real matrices).

Problem 6

Hint 1. Let $\omega = e^{2\pi i/3}$ and $S = A + \omega B$; since A, B are real, $\bar{S} = A + \bar{\omega}B$. Expand $S\bar{S}$ and use $A^2 + B^2 = AB$ and $1 + \bar{\omega} = -\omega$.

Hint 2. $S\bar{S} = AB + \bar{\omega}AB + \omega BA = \omega(BA - AB)$. Take determinants: $|\det S|^2 = \omega^n \det(BA - AB)$. The left-hand side is real and, as $BA - AB$ is invertible, non-zero; so ω^n is real, i.e. $3 \mid n$.

Problem 7

Hint 1. (a) The polynomial $x^3 - x - 1$ has a real root r; take $A = rE$.

Hint 2. (b) The eigenvalues of A are roots of $x^3 - x - 1$. It has exactly one real root, and it is positive ($f(0) = -1 < 0$, and f is negative on $(-\infty, 0]$ because its maximum there is $f(-\frac{1}{\sqrt{3}}) < 0$). The non-real eigenvalues of the real matrix A come in conjugate pairs, so $\det A = r^a|z|^{2b} > 0$.

Problem 8

Hint 1. (a) $v = \frac{v+Av}{2} + \frac{v-Av}{2}$, and the two summands are eigenvectors for 1 and -1 (or zero). So $V = \ker(A - E) \oplus \ker(A + E)$.

Hint 2. (b) Commuting diagonalisable operators are simultaneously diagonalisable: each preserves the eigenspaces of the others. In a common eigenbasis every involution is $\mathrm{diag}(\pm 1, \ldots, \pm 1)$. Answer: 2^n.

Problem 9

Hint 1. In block form the matrix is $M = \begin{pmatrix} N & -E \\ E & N \end{pmatrix}$ with $N = \begin{pmatrix} a & -a \\ a & -a \end{pmatrix}$, $N^2 = 0$. Its characteristic polynomial is $\det\big((x - N)^2 + E\big) = (x^2 + 1)^2$: the eigenvalues are $\pm i$, each twice.

Hint 2. If $M = B^2$ with B rational, every eigenvalue μ of B has $\mu^2 = \pm i$, so $\mu^4 = -1$. The polynomial $x^4 + 1$ is irreducible over $\mathbb{Q}$, hence the characteristic polynomial of B is $x^4 + 1$. Its roots are distinct, so B is diagonalisable, and so is $M = B^2$.

Hint 3. A diagonalisable M with eigenvalues $\pm i$ satisfies $M^2 = -E$. But $M^2 = \begin{pmatrix} -E & -2N \\ 2N & -E \end{pmatrix}$, so $N = 0$ and $a = 0$. Conversely, for $a = 0$ the matrix M is the square of the companion matrix of $x^4 + 1$ (the matrix of multiplication by $\zeta = e^{\pi i/4}$ on $\mathbb{Q}(\zeta)$ in the basis $1, \zeta, \zeta^2, \zeta^3$; its square is multiplication by i). Answer: $a = 0$.

Problem 10

Hint 1. The minimal polynomial $p(x) = x^n + x - 1$ has degree n, so it is also the characteristic polynomial. Check that p has n distinct roots: a common root of p and $p' = nx^{n-1} + 1$ would satisfy $\lambda = \frac{n}{n-1} > 0$ and $\lambda^{n-1} = -\frac{1}{n} < 0$.

Hint 2. So A is diagonalisable with eigenvalues $\lambda_1, \ldots, \lambda_n$, the matrix $nA^{n-1} + E = p'(A)$ is invertible, and the trace equals $\sum_i \frac{\lambda_i^{n-2}}{p'(\lambda_i)}$.

Hint 3. For a polynomial q of degree $\leqslant n - 2$, Lagrange interpolation at the roots of p gives $q(x) = \sum_i q(\lambda_i) \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j}$; comparing the coefficients of x^{n-1} gives $\sum_i \frac{q(\lambda_i)}{p'(\lambda_i)} = 0$. Take $q(x) = x^{n-2}$.

Problem 11

Hint 1. $\operatorname{rank} A \neq \operatorname{rank} A^2$ means $\ker A \subsetneq \ker A^2$: there is x with $A^2x = 0$ but $y = Ax \neq 0$. Look for B of rank one, $B = yz^T$.

Hint 2. Take $z \neq 0$ with $z^T A = 0$ (it exists since A is singular). Then $AB = (Ay)z^T = O$, $BA = y(z^T A) = O$ and $B^2 = (z^T y)B$ with $z^T y = z^T Ax = 0$.

Problem 12

Hint 1. Prove the stronger claim: a non-scalar A is similar to a matrix with diagonal $(0, \dots, 0, \operatorname{tr} A)$. Some v is not an eigenvector, so v, Av are independent; in a basis starting with v, Av the first column is $(0, 1, 0, \dots, 0)^T$, so the $(1,1)$ entry is 0.

Hint 2. Let A' be the lower-right $(n-1) \times (n-1)$ block. Conjugating by $\operatorname{diag}(1, P)$ keeps the first column and replaces A' by $P^{-1}A'P$; so if A' is not scalar, the induction hypothesis finishes the job.

Hint 3. If $A' = cE$ ($n \geqslant 3$), replace the third basis vector w_3 by $w_3 + v$: the first column does not change, and A' gets a non-zero off-diagonal entry. For $n = 2$ the basis v, Av already gives diagonal $(0, \operatorname{tr} A)$.

Problem 13

Hint 1. C is nilpotent if and only if $\chi_C(x) = x^n$. Each coefficient of $\det(xE - A - tB)$ is a polynomial in t of degree at most n.

Hint 2. These polynomials vanish at $n+1$ points (except the leading x^n), so they vanish identically: $A + tB$ is nilpotent for all t. Hence $(A + tB)^n = O$ for all t. Look at the coefficients of t^0 and t^n.

Problem 14

Hint 1. The coefficients of χ_A are, up to sign, the sums of principal minors, so $\chi_A = x^n$ and $A^n = O$ by Cayley–Hamilton. For the permutation, consider the directed graph with an edge $j \to k$ whenever $a_{jk} \neq 0$.

Hint 2. It suffices to show this graph has no directed cycles: then a topological order of the vertices gives σ. Diagonal entries are 1×1 minors, so there are no loops.

Hint 3. Take a shortest directed cycle $j_1 \to \dots \to j_m \to j_1$. A chord would create a shorter cycle, so the induced subgraph on these vertices is exactly this cycle. In the expansion of the principal minor on $\{j_1, \dots, j_m\}$ only one permutation contributes. So the minor is $\pm \prod a_{j_i j_{i+1}} \neq 0$, a contradiction.

Problem 15

Hint 1. (a) Pass to the Jordan form of X. If X is diagonalisable, take cube roots of the eigenvalues of X^2. Otherwise $X = \lambda E + N$ with $N \neq O$, $N^2 = O$, and $X^2 = \lambda^2 E + 2\lambda N$.

Hint 2. For $\lambda = 0$ take $Y = O$. For $\lambda \neq 0$ look for $Y = cE + dN$: $Y^3 = c^3 E + 3c^2 dN$, so solve $c^3 = \lambda^2$, $3c^2 d = 2\lambda$.

Hint 3. (b) Take the nilpotent 3×3 Jordan block A: $A^2 \neq O$, $A^3 = O$. If $Z^3 = A^2$, then $Z^6 = O$, so Z is nilpotent, and in size 3 this forces $Z^3 = O \neq A^2$.

Problem 16

Hint 1. Compute A^2. Its diagonal entries are $n-1$, and its off-diagonal entries are the common scalar product s of two rows: $A^2 = (n - 1 - s)E + sJ$, where J is the all-ones matrix. Note that $s \leqslant n - 2$.

Hint 2. If $s = 0$, all $|\lambda_i| = \sqrt{n-1}$ and $\sum |\lambda_i| = n\sqrt{n-1} > 2n - 2$. If $s \neq 0$, the line spanned by $\mathbf{1} = (1, \dots, 1)$ is an eigenspace of A^2 of its own, and A commutes with A^2, so $A\mathbf{1} = r\mathbf{1}$: all row sums are equal to some r, with $r^2 = (n-1)(1+s)$. The other eigenvalues satisfy $\lambda^2 = n - 1 - s$.

Hint 3. Eliminate s: with $t = |r| \in [0, n-1]$,

$$\sum |\lambda_i| = t + (n-1)\sqrt{n-1-s} = t + \sqrt{n-1}\,\sqrt{n(n-1)-t^2}.$$

This is a concave function of t, so its minimum is at an end of the interval: $(n-1)\sqrt{n} > 2n-2$ at $t = 0$ (here $n \geqslant 5$ is used), and exactly $2n-2$ at $t = n-1$.

Hint 4. Equality forces $|r| = n-1$: in every row all off-diagonal entries have the same sign, and since all row sums are equal, it is the same sign for all rows. Answer: equality holds exactly for $A = \pm(J - E)$.